Declan Chan

declan.chan@mail.utoronto.ca • declanwhchan.github.io • declanwhchan

Summary

Engineering Science student at the University of Toronto with experience in machine learning research, aerospace systems, CAD design, and software development.

Skills and Proficiencies

Programming: Python, C, MATLAB, HTML/CSS, Git

Design: SolidWorks, OnShape

Education

Engineering Science – University of Toronto *2025–Present*
- 2× Dean's Honours List

International Baccalaureate Diploma Programme *2021–2025*
- IB Score: 45/45
- OSSD Top 6: 99.67%
- Director's Achievement Award

Experiences

Machine Learning Researcher – AutoDIAL *2026–Present*
- Autonomous Discovery of Alloys (AutoDIAL) researcher.
- Performed adversarial attacks on Machine Learning Force Field models to test their robustness.
- Benchmarked the failure points of Message Passing Atomic Cluster Expansion training models.
- Implemented testing workflows using Python, PyTorch, and Jupyter Notebook.

Space Systems Engineer – University of Toronto Aerospace Team *2026–Present*
- Contributed to CubeSat attitude determination and control systems.
- Simulated magnetic environments using a Helmholtz cage.

Senior Karate Instructor – Northern Karate Schools *2021–Present*
- Yondan: Certified 4th Degree Black Belt.
- Certification: Blended Standard First Aid & CPR/AED Level C.
- Gold medal at the Toronto International Tournament of Martial Arts Champions.
- Teach 100+ kids, teens, and adults every week alongside private classes.

Mechanical Design Engineer – Robotics for Space Exploration *2025–2026*
- Designed robotic arm components using CAD for 3D modelling.
- Performed Finite Element Analysis using SolidWorks Simulation to evaluate designs.

NFT Automation *2021–2022*
- Programmed a macro to create NFTs on OpenSea.io with Ethereum.
- Explored the Polygon Ethereum blockchain scaling solution.

Technical Projects

GenAI Genesis Hackathon *2026*
- Developed an interactive platform visualizing human disease progression as a team of four.
- Implemented a RAG system trained on PubMed data via IBM watsonx.ai.
- Leveraged Agentic AI with React and three.js to build a 3D web application.

Civil Engineering Bridge Competition *2025*
- Optimized structural performance and prototyped a matboard bridge using Python.
- Placed 2nd out of 75 teams in Engineering Science as a team of four.

Yarkovsky Effect Orbital Analysis *2024–2025*
- Explored the existence of the Yarkovsky effect from asteroids' physical and orbital properties.
- Conducted research and physics calculations under teacher mentorship.

Hydrolysis Investigation *2024–2025*

- Investigated the effect of temperature on the rate constant in the hydrolysis of ethyl acetate.
- Collected data by conducting experiments using a personalized lab methodology.

Rotational Dynamics Study *2023–2024*

- Studied how the velocity of a volleyball serve affects the ball's angular velocity.
- Modelling the Magnus effect through calculations and experimental data.

Violin Modelling *2023–2024*

- Analyzed how the density of a violin affects its market price.
- Used integration to model and calculate volumes of irregular shapes.

Leadership & Activities

Varsity Volleyball *2022–2025*

- Team Captain in Grade 10 and 12.
- Tier 1 YRAA Championship junior varsity second place.
- Tier 1 YRAA Championship senior varsity second place.

Rubik's Cube Competitions *2022–2025*

- Top 90 in Canada for 3x3x3 fewest moves.
- Volunteered as a judge for the World Cube Association.
- Club President who taught beginners and organized school-wide competitions.

Music Performance *2014–2023*

- Theory and Instrumental: Level 10 RCM comprehensive certificate (First Class Honors).
- Level 10 RCM practical piano certificate (First Class Honors with Distinction).
- Level 8 RCM practical violin certificate (First Class Honors with Distinction).

Rep Volleyball *2019–2023*

- Team Captain and 2× MVP.
- Silver medal in Canada's 2023 Youth Nationals.